

Figure 12: Parented

- **Envelope:** Again, this is more a feature than a display mode, but this visualisation will be useful to tweak your rig later. It lets you easily tell which part of your character this bone will animate, and it is visually possible to change the zone of influence exclusively in this display mode. The zone is only visible in *Edit* mode or *Pose* mode, though.

Creating the skeleton for the goldfish

Shift to side view (press 3). Click at the back of the fish, just before where the tail should begin. Press space-bar; click the *Add* menu's *Armature* option to add a default parent bone that will act as the hip bone. It's easiest to create all the bones first, parent them in the correct hierarchy, and later name them appropriately. Care has to be taken to see that the first bone is properly created and placed in the body of the fish—ensure that the X-ray option is on at the bone creation level. Look at Figures 6 to 11 and ensure your bone is aligned the same way. Click the tip of the hip bone in the edit mode. Press E to extrude a single bone, which will be the abdomen bone. Extrude another two, for the chest and head. Similarly, add bones for fins, the tail and the jaw. Parent the bones as seen in the figures: RMB-click-select the bone in *Edit* mode; then shift-select the second bone to which you want to parent the first bone. Press **Ctrl+P** and from the pop-up, select **Keep Offset**. In this way, parent all the bones appropriately.

Parenting the body to the skeleton

Once the skeleton is created, parent the fish body to the skeleton. First select the body and then shift-select the skeleton. Press **Ctrl+P** to bring up the *Make Parent to—Armature/ Object* dialogue box. Make the body a parent to the skeleton by selecting the *Armature* option. This brings up another pop-up dialogue asking whether

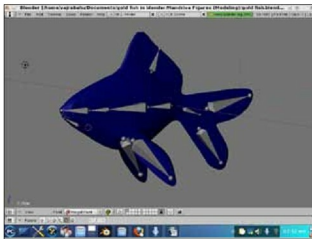


Figure 13: Weight painting mode

you wish to create vertex groups. Choose the third option, *Create from closest bones*. This is how the fish is rigged. The basic deformations on the fish could be tested by going into the pose mode and rotating the bones on either end of the fish. You will observe that the corresponding body part also moves. For a better deformation, you could even switch to the *autoIK* mode in the armature panel. Figure 12 shows the dotted lines that result from parenting all the parts into a single working skeleton.

Painting weights for exact deformations

Once the fish body is parented to the armature, all the parts of the fish may not deform the way you want. For exact deformations, slight alterations have to be made, in terms of the influence of a particular bone on a particular part of the body. To paint weights, get into weight painting mode, using the drop-down menu from the separator bar.

Painting with the slider at the highest level paints red, which stands for full bone influence; the slider paints green at the middle level—half the bone influence; the slider's extreme left provides zero bone influence. Paint weights if you are not happy with the deformations that take place in the pose mode. Figure 13 shows the default weight painting mode.

In the next article in this series on Blender, we will be creating controls for our fish, and will observe ways of animating it. **END**

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